Square anisotropy: the Wulff square is never fractional-isoperimetric

A sharp partial result for the open problem in arXiv:1304.0699

Autonomous proof audit — lane 9

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Candidate partial result. For every $0 < s < 1$, the square unit ball is not a minimizer of its own anisotropic fractional perimeter: its nonlocal mean curvature varies strictly along each side. The conclusion transfers affinely to every parallelogram. In contrast, the square is the unique minimizer among all fixed-area aligned rectangles. Thus the true optimizer must leave the most obvious Wulff family.

1 Source problem and current frontier

For an origin-symmetric convex body $K \subset \mathbb{R}^n$ and $0 < s < 1$, Ludwig defines

$$P_s(E,K) = \int_E \int_{E^c} \frac{dx\,dy}{\|x-y\|_K^{n+s}}.$$

The anisotropic fractional isoperimetric problem is to determine the equality cases in

$$P_s(E,K) \geq \gamma_s(K)|E|^{(n-s)/n}.$$

The source explicitly says that determination of the minimizers remains open.

The Euclidean case is solved by balls, and an invertible linear change of variables solves every ellipsoidal anisotropy. For nonellipsoidal K, later work proves structural results but not a classification: Kreuml proves an unconditional star-body conclusion under strict convexity and unconditionality [2], while Cai derives a necessary fractional-area measure condition for convex optimizers [3]. The result below gives an explicit all-s obstruction for the most elementary crystalline anisotropy.

2 The Wulff square is not stationary

Let

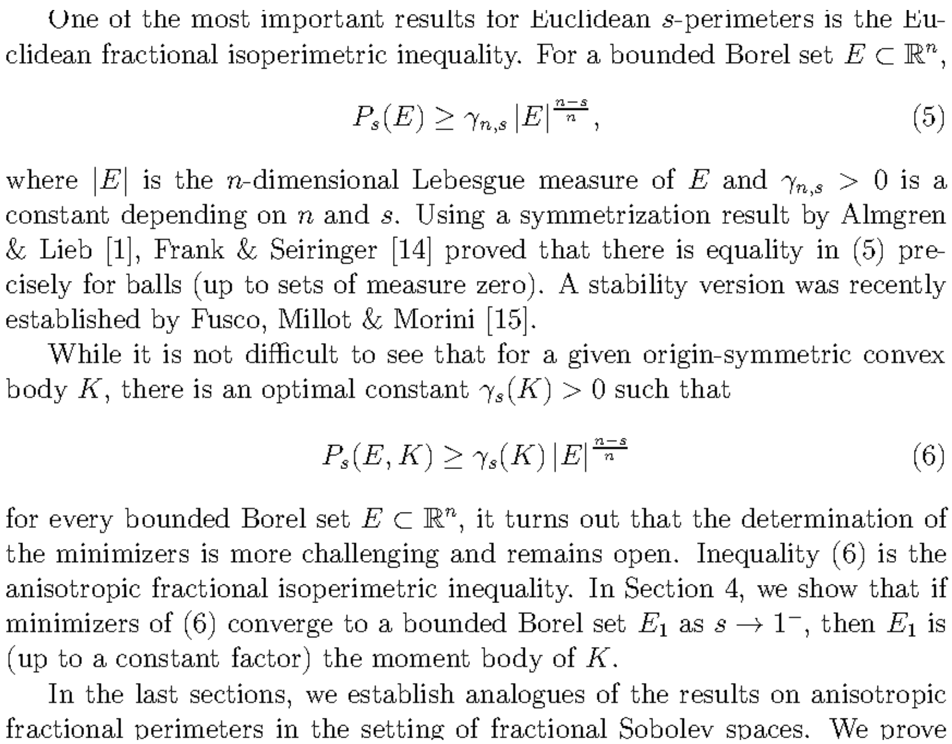
$$Q = [-1,1]^2, \qquad \|z\|_Q = \max\{|z_1|,|z_2|\}, \qquad k_s(z) = \|z\|_Q^{-2-s}.$$

At the regular boundary point $p_t = (1,t)$, $|t| < 1$, define the principal-value nonlocal curvature

$$H_s(t) = \mathrm{PV}\int_{\mathbb{R}^2} \big(\mathbf{1}_{Q^c}(y) - \mathbf{1}_Q(y)\big) k_s(y-p_t)\,dy.$$

Lemma 1. *The function H_s is even and is strictly increasing on $(0,1)$. More precisely,*

$$H_s'(t) = 2\int_{-2}^{0} \big[\max\{|x|,1-t\}^{-2-s} - \max\{|x|,1+t\}^{-2-s}\big]\,dx > 0. \tag{1}$$



One of the most important results for Euclidean s-perimeters is the Euclidean fractional isoperimetric inequality. For a bounded Borel set $E \subset \mathbb{R}^n$,

$$P_s(E) \geq \gamma_{n,s}\,|E|^{\frac{n-s}{n}}, \tag{5}$$

where $|E|$ is the n-dimensional Lebesgue measure of E and $\gamma_{n,s} > 0$ is a constant depending on n and s. Using a symmetrization result by Almgren & Lieb [1], Frank & Seiringer [14] proved that there is equality in (5) precisely for balls (up to sets of measure zero). A stability version was recently established by Fusco, Millot & Morini [15].

While it is not difficult to see that for a given origin-symmetric convex body K, there is an optimal constant $\gamma_s(K) > 0$ such that

$$P_s(E, K) \geq \gamma_s(K)\,|E|^{\frac{n-s}{n}} \tag{6}$$

for every bounded Borel set $E \subset \mathbb{R}^n$, it turns out that the determination of the minimizers is more challenging and remains open. Inequality (6) is the anisotropic fractional isoperimetric inequality. In Section 4, we show that if minimizers of (6) converge to a bounded Borel set E_1 as $s \to 1^-$, then E_1 is (up to a constant factor) the moment body of K.

In the last sections, we establish analogues of the results on anisotropic fractional perimeters in the setting of fractional Sobolev spaces. We prove

Figure 1: The open minimizer problem in arXiv:1304.0699, printed page 4.

Proof. After the change of variables $z = y - p_t$, the translated interior is

$$R_t = Q - p_t = [-2, 0] \times [-1 - t, 1 - t].$$

In the difference $H_s(t) - H_s(0)$ the singular half-space contributions cancel, and hence

$$H_s(t) - H_s(0) = -2 \int_{\mathbb{R}^2} (\mathbf{1}_{R_t} - \mathbf{1}_{R_0})(z) k_s(z)\, dz.$$

The symmetric difference stays away from the origin to first order, so this finite expression may be differentiated by moving the two horizontal edges. If

$$J(t) = \int_{-2}^{0} \int_{-1-t}^{1-t} k_s(x, y)\, dy\, dx$$

is interpreted only through differences, then

$$J'(t) = \int_{-2}^{0} \big(k_s(x, 1 + t) - k_s(x, 1 - t)\big)\, dx.$$

Since $H_s' = -2J'$, this is (1). The integrand there is nonnegative and is strictly positive, for example, whenever $|x| < 1 - t$. Reflection in the horizontal axis gives evenness. $\square$

Theorem 1. *For every $0 < s < 1$, no translate or dilate of Q minimizes $P_s(\,\cdot\,, Q)$ under a fixed-area constraint. More generally, if K is any origin-symmetric parallelogram, no homothetic copy of K is a minimizer of $P_s(\,\cdot\,, K)$.*

Proof. A volume-constrained minimizer is stationary under smooth normal variations supported in the relative interior of its sides; its nonlocal curvature must therefore equal one Lagrange multiplier there. Here this necessary condition can also be used constructively. Choose two small side intervals, one near $t = 0$ and one near a fixed $t_0 \in (0, 1)$, and a smooth function φ supported on them with zero integral, positive near 0, and negative near t_0, with equal masses. Replace the right boundary $x = 1$ by $x = 1 + \varepsilon\varphi(t)$. The area is unchanged exactly, and the standard first-variation formula gives

$$\frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} P_s(E_\varepsilon, Q) = \int_{-1}^{1} H_s(t)\varphi(t)\, dt < 0$$

by Lemma 1. Thus Q is not even stationary.

If A is invertible, then

$$\|A(x-y)\|_{AQ} = \|x-y\|_Q, \quad P_s(AE, AQ) = |\det A|^2 P_s(E,Q), \quad |AE| = |\det A|\,|E|.$$

Every origin-symmetric parallelogram is AQ for some A, so the nonminimality transfers. Translation and dilation invariance give the stated homothetic version. □

3 Nevertheless, the square is best among rectangles

The previous theorem is not a mere aspect-ratio instability. Let

$$R_q = [-q/2, q/2] \times [-1/(2q), 1/(2q)], \qquad q \geq 1,$$

so every R_q has area one.

Theorem 2. *For every $0 < s < 1$, the map $q \mapsto P_s(R_q, Q)$ is strictly increasing on $(1, \infty)$. Hence the square R_1 is the unique minimizer, up to interchanging the axes, among all area-one axis-aligned rectangles.*

Proof. For a rectangle with side lengths $A \geq B$, its covariogram is

$$g_{A,B}(z) = (A - |z_1|)_+(B - |z_2|)_+.$$

The translation formula and the layer-cake identity $r^{-2-s} = (2+s)\int_r^\infty t^{-3-s}\, dt$ give

$$P_s(R, Q) = (2+s)\int_0^\infty t^{-3-s}\big(4ABt^2 - h_A(t)h_B(t)\big)\, dt, \tag{2}$$

where

$$h_C(t) = \begin{cases} 2Ct - t^2, & 0 \leq t \leq C, \\ C^2, & t \geq C. \end{cases}$$

Indeed, the bracket in (2) is $\int_{[-t,t]^2}(|R| - g_{A,B}(z))\, dz$.

Substitute $A = q$, $B = q^{-1}$, split the integral at B and A, and integrate the resulting powers. Direct simplification yields

$$\frac{d}{dq}P_s(R_q, Q) = \frac{2(s+2)q^{-s-3}}{s(1-s)(1+s)}G_s(q),$$
$$G_s(q) = 2sq^{2+2s} - (s+1)q^{2s} + 1 - s.$$

Now $G_s(1) = 0$ and

$$G_s'(q) = 2s(s+1)q^{2s-1}(2q^2 - 1) > 0 \qquad (q \geq 1).$$

The claimed strict increase follows. □

Theorem 2 and Theorem 1 together force any true optimizer for square anisotropy to be a genuinely nonrectangular shape. Affine covariance gives the analogous statement for parallelogram anisotropy and aligned parallelogram competitors.

4 Upgrade audit, scope, and novelty

Several routes toward a full classification were tested. Ellipsoids reduce to the Euclidean theorem, but anisotropic Riesz rearrangement is unavailable outside that case. Steiner symmetrization reaches unconditional star bodies only under additional hypotheses. The $s \uparrow 1$ limit selects the moment body but does not identify any fixed-s optimizer. Finally, Lemma 1 turns the square problem into a nonlinear constant nonlocal-curvature equation for an unknown rounded box body; neither the rectangle calculation nor current Alexandrov-type theorems for radial kernels control that anisotropic equation. These are genuine obstructions, so no full classification is claimed.

Bounded searches using the exact source phrase, title, square/cube, parallelogram, Wulff shape, optimizer, and anisotropic nonlocal curvature found the source, Kreuml's structural theorem, and Cai's recent necessary condition, but no all-s square nonstationarity calculation or exact rectangle comparison. Novelty is therefore plausible but unverified. Expert review should focus on the first-variation step at a polygonal boundary and the layer-cake constants.

References

[1] M. Ludwig, *Anisotropic fractional perimeters*, J. Differential Geom. **96** (2014), 77–93; arXiv:1304.0699.

[2] A. Kreuml, *The anisotropic fractional isoperimetric problem with respect to unconditional unit balls*, Commun. Pure Appl. Anal. **20** (2021), 783–799, doi:10.3934/cpaa.2020290.

[3] X. Cai, *Anisotropic fractional area measures*, arXiv:2510.05279 (2025), Theorem 6.1.